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PSAC Exam Practice — Mauritius

# 2026

## Grade 6 • Science

**Specimen Paper 4**

Time allowed: **1 hour 45 minutes**

Total marks: **100**

Candidate Name: \_\_\_\_\_

School: \_\_\_\_\_

Date: \_\_\_\_\_

Index No.: \_\_\_\_\_

### Instructions to Candidates

1. Write your name, school and index number clearly on this paper.
2. Answer **ALL** questions in **ALL FOUR** sections.
3. Write your answers in the spaces provided on this paper.
4. All diagrams must be neat, clearly labelled and drawn in pencil.
5. The marks for each question are shown in square brackets [ ].
6. Calculators and electronic devices are **NOT** permitted.
7. Where working is required, show all steps clearly.

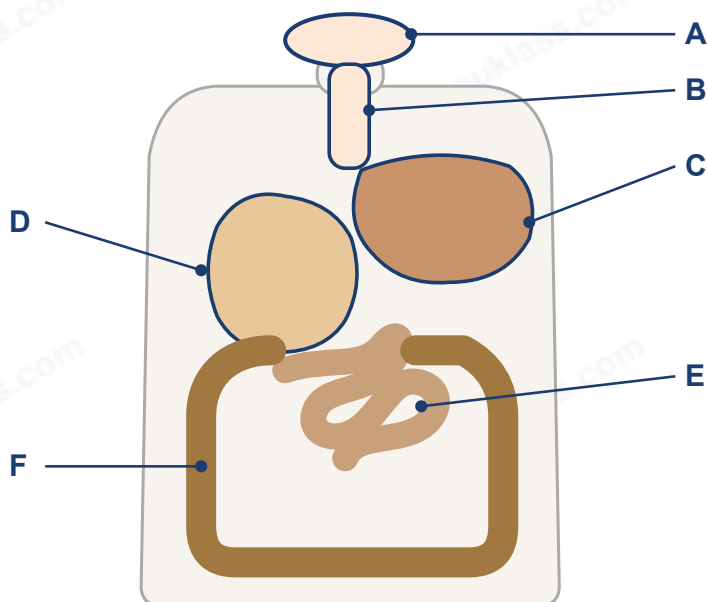
| Section | Topic                                  | Available | Awarded |
|---------|----------------------------------------|-----------|---------|
| A       | Living Things — Human Body Systems     | 30        |         |
| B       | Matter — Dissolving & Chemical Changes | 25        |         |
| C       | Energy — Electricity & Circuits        | 25        |         |
| D       | Earth & Space — Weather & Climate      | 20        |         |

| Section | Topic | Available | Awarded |
|---------|-------|-----------|---------|
| TOTAL   |       | 100       |         |

## SECTION A — Living Things: Human Body Systems [30 marks]

- 1 The diagram below shows the human digestive system. Study it and answer the questions.

[10]



- (a) Name each labelled part A–F of the digestive system. [6]

|          |          |
|----------|----------|
| A: _____ | B: _____ |
| C: _____ | D: _____ |
| E: _____ | F: _____ |

- (b) Where is most digestion and absorption of nutrients into the blood carried out?

[1]

- (c) What is the role of the liver in digestion?

[1]

- (d) Name the **three** types of food nutrient that are digested (broken down) in the digestive system.

[2]

1. \_\_\_\_\_ 2. \_\_\_\_\_ 3. \_\_\_\_\_
- \_\_\_\_\_

2 This question is about the circulatory system.

[10]

- (a) Name the three main parts of the circulatory system and state the function of each.

[6]

| Part | Function |
|------|----------|
|      |          |
|      |          |
|      |          |

- (b) What is the difference between an *artery* and a *vein*?

[2]

- (c) Regular exercise makes the heart stronger. Describe **one** other benefit of regular exercise for the body.

[1]

- (d) Name the gas carried by red blood cells from the lungs to all body cells.

[1]

3 This question is about the respiratory system and healthy living.

[10]

- (a) Describe what happens in the lungs during breathing in and breathing out. Use the terms *diaphragm*, *ribcage*, *oxygen*, and *carbon dioxide* in your answer.

[4]

(b) State the **two** gases exchanged in the lungs during gas exchange. [2]

Gas that passes from the air into the blood: \_\_\_\_\_

Gas that passes from the blood into the air: \_\_\_\_\_

(c) Smoking damages the lungs. Name **two** diseases that can be caused by smoking. [2]

1. \_\_\_\_\_ 2. \_\_\_\_\_

(d) State **two** components of a balanced diet that provide the body with energy. [2]

1. \_\_\_\_\_ 2. \_\_\_\_\_

## SECTION B — Matter: Dissolving & Chemical Changes [25 marks]

- 4 This question investigates how quickly sugar dissolves in water. [9]

A pupil dissolves the same mass of sugar in water under different conditions and times how long it takes. Results are shown below.

| Experiment | Temperature (°C) | Stirring? | Sugar grain size | Time to dissolve (s) |
|------------|------------------|-----------|------------------|----------------------|
| 1          | 20               | No        | Large            | 240                  |
| 2          | 60               | No        | Large            | 85                   |
| 3          | 20               | Yes       | Large            | 120                  |
| 4          | 20               | No        | Small (crushed)  | 95                   |

- (a) Compare experiments 1 and 2. What conclusion can you draw about the effect of temperature on dissolving? Explain why. [2]

- (b) Compare experiments 1 and 4. State what variable was changed and what effect it had on the rate of dissolving. [2]

- (c) In this investigation, state **two** variables that were kept the same (controlled) to make it a fair test. [2]

1. \_\_\_\_\_
2. \_\_\_\_\_

- (d) Predict the outcome if experiment 5 used hot water (60°C), stirring AND small (crushed) sugar. Would it dissolve faster or slower than experiments 2, 3, or 4? Explain.

[2]

(e) What does it mean when a solution is *saturated*?

[1]

5 This question is about chemical changes and acids and bases.

[8]

(a) Classify each change below as *physical* or *chemical*. Give one reason for each answer.

[4]

| Change                                      | Physical or Chemical? | Reason |
|---------------------------------------------|-----------------------|--------|
| Dissolving salt in water                    |                       |        |
| Vinegar reacting with baking soda (fizzing) |                       |        |

(b) Litmus indicator is added to three unknown liquids. The results are shown below. Identify each liquid as acid, alkali, or neutral.

[3]

| Liquid            | Litmus colour      | Acid, Alkali or Neutral? |
|-------------------|--------------------|--------------------------|
| Lemon juice       | Red                |                          |
| Washing-up liquid | Blue               |                          |
| Pure water        | Purple (unchanged) |                          |

(c) A pupil adds baking soda (an alkali) to lemon juice (an acid) and measures the pH. The pH changes from 3 to 7. What term describes this process?

[1]

6 This question is about the properties and uses of materials.

[8]

- (a) Explain, with an example, what a *composite material* is and why composites are useful.

[3]

- (b) Complete the table by selecting the best material from the list below for each purpose, and giving one relevant property.  
*Choose from: copper, wood, polystyrene, glass, stainless steel*

[5]

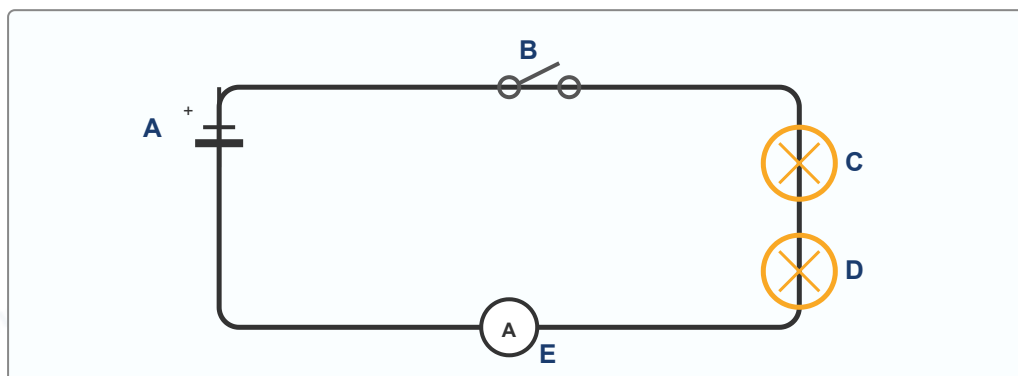
| Purpose                             | Best Material | Relevant Property |
|-------------------------------------|---------------|-------------------|
| Electrical wiring in a house        |               |                   |
| A disposable cup to keep drinks hot |               |                   |
| A kitchen knife blade               |               |                   |
| A magnifying lens                   |               |                   |
| A chair seat                        |               |                   |



## SECTION C — Energy: Electricity & Circuits [25 marks]

- 7 The diagram below shows an electrical circuit. Study it carefully and answer the questions.

[9]



- (a) Identify the components labelled A–E in the circuit diagram. [5]

A: \_\_\_\_\_ B: \_\_\_\_\_  
C: \_\_\_\_\_ D: \_\_\_\_\_  
E: \_\_\_\_\_

- (b) The switch B is currently open. What will happen to the bulbs? Explain why.

[2]

- (c) The two bulbs C and D are connected in *series*. What would happen to bulb D if bulb C blew (broke)? Explain why.

[2]

- 8 This question compares series and parallel circuits.

[9]

- (a) State **two** differences between a series circuit and a parallel circuit. [4]

- (b) Christmas tree lights are connected in parallel. Explain why this is better than connecting them in series. [2]

- (c) Name **three** materials that are good conductors of electricity. [3]

1. \_\_\_\_\_ 2. \_\_\_\_\_ 3. \_\_\_\_\_  
\_\_\_\_\_

- 9 This question is about electrical safety and magnetism. [7]

- (a) State **three** rules for electrical safety in the home. [3]

1. \_\_\_\_\_  
2. \_\_\_\_\_  
3. \_\_\_\_\_

- (b) A fuse is a safety device in an electrical circuit. Explain how a fuse protects appliances from damage. [2]

- (c) Describe how you could make a simple electromagnet using a battery, wire and an iron nail. State **one** use of an electromagnet.

[2]



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**SECTION D — Earth & Space: Weather & Climate [20 marks]**

**10** This question is about weather and climate.

[10]

**(a)** Explain the difference between *weather* and *climate*.

[2]

**(b)** The table below shows weather data recorded in two Mauritius towns over one week. Use the data to answer the questions.

| Town       | Average temp (°C) | Total rainfall (mm) | No. of sunny days |
|------------|-------------------|---------------------|-------------------|
| Port Louis | 28                | 12                  | 5                 |
| Curepipe   | 22                | 48                  | 2                 |

**(i)** Which town is cooler and wetter? Suggest a reason related to its location.

[2]

**(ii)** What is the difference in average temperature between the two towns?

[1]

**(c)** What is a *cyclone*? Describe **two** effects a cyclone can have on Mauritius.

[3]

- (d) State **two** ways in which humans are contributing to climate change (global warming). [2]

1. \_\_\_\_\_
2. \_\_\_\_\_

11 This question is about climate zones and the environment. [10]

- (a) Name the **three** main climate zones of the Earth and state one characteristic of each. [3]

| Climate Zone | One Characteristic |
|--------------|--------------------|
|              |                    |
|              |                    |
|              |                    |

- (b) How does the greenhouse effect work? Use the terms *Sun*, *greenhouse gases*, *infrared radiation*, and *atmosphere* in your explanation. [4]

- (c) Name **two** renewable energy sources that can replace fossil fuels and reduce greenhouse gas emissions. [2]

1. \_\_\_\_\_ 2. \_\_\_\_\_

- (d) Mauritius is an island nation. Name **one** specific risk that rising sea levels pose to Mauritius.

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**END OF PAPER 4** — Check all your answers carefully before handing in.

## Mark Scheme — Specimen Paper 4 (2026) • Grade 6 Science

**For teachers and parents:** Accept any answer that demonstrates equivalent scientific understanding. Where wording differs, award the mark if the key concept is clearly expressed. Answers in *italics* are alternative acceptable responses. This paper is for practice purposes only. Produced by Noukclass.com.

| Q | Part | Acceptable Answer(s)                                                                                                                                                                                                                                                                | Marks |
|---|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1 | (a)  | A=Mouth, B=Oesophagus (gullet), C=Liver, D=Stomach, E=Small intestine, F=Large intestine. [1 each]                                                                                                                                                                                  | 6     |
|   | (b)  | Small intestine                                                                                                                                                                                                                                                                     | 1     |
|   | (c)  | The liver produces bile (which helps break down fats / emulsify fats). It also processes absorbed nutrients and removes toxins from the blood.                                                                                                                                      | 1     |
|   | (d)  | Carbohydrates, proteins, fats (lipids). [1 each]                                                                                                                                                                                                                                    | 2     |
| 2 | (a)  | Heart: pumps blood around the body. Blood vessels (arteries/veins/capillaries): carry blood to all parts of the body. Blood: transports oxygen, nutrients, and waste. [2 per correct pair, max 6]                                                                                   | 6     |
|   | (b)  | Arteries: carry blood away from the heart; have thick muscular walls; carry oxygenated blood (except pulmonary artery). Veins: carry blood towards the heart; thinner walls; have valves; carry deoxygenated blood (except pulmonary vein). [1 per correct difference, max 2]       | 2     |
|   | (c)  | Any one: strengthens muscles; helps maintain healthy weight; improves flexibility; strengthens bones; reduces risk of disease; improves mental health.                                                                                                                              | 1     |
|   | (d)  | Oxygen                                                                                                                                                                                                                                                                              | 1     |
| 3 | (a)  | Breathing in: diaphragm contracts (moves down), ribcage moves up and out, chest volume increases, air containing oxygen enters the lungs. Breathing out: diaphragm relaxes (moves up), ribcage moves down and in, air containing carbon dioxide leaves. [1 per correct step, max 4] | 4     |
|   | (b)  | Into blood: oxygen. Into air: carbon dioxide. [1 each]                                                                                                                                                                                                                              | 2     |
|   | (c)  | Any two: lung cancer, emphysema, chronic bronchitis, heart disease. [1 each]                                                                                                                                                                                                        | 2     |
|   | (d)  | Any two: carbohydrates (starch/sugars), fats. [1 each]                                                                                                                                                                                                                              | 2     |
| 4 | (a)  | Higher temperature causes sugar to dissolve faster [1]; heat gives water molecules more                                                                                                                                                                                             | 2     |

| Q | Part | Acceptable Answer(s)                                                                                                                                                                                                                                                                                                                                  | Marks |
|---|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|   |      | energy so they move faster and break down the sugar particles more quickly [1].                                                                                                                                                                                                                                                                       |       |
|   | (b)  | Variable changed: grain size (large vs crushed small) [1]. Effect: smaller grains dissolve faster (95s vs 240s) because they have a larger surface area in contact with water [1].                                                                                                                                                                    | 2     |
|   | (c)  | Any two from: same mass of sugar, same volume of water, same type of sugar, same size of container. [1 each]                                                                                                                                                                                                                                          | 2     |
|   | (d)  | Faster than any of experiments 2, 3 or 4 [1]; because all three factors that speed up dissolving (temperature, stirring, small particle size) are combined [1].                                                                                                                                                                                       | 2     |
|   | (e)  | A solution is saturated when no more solute can dissolve in the solvent at that temperature.                                                                                                                                                                                                                                                          | 1     |
| 5 | (a)  | Dissolving salt: physical; it is reversible (the salt can be recovered by evaporation / no new substance is formed). Vinegar + baking soda: chemical; a new substance (carbon dioxide gas) is formed and the change is irreversible. [2 per row]                                                                                                      | 4     |
|   | (b)  | Lemon juice: acid. Washing-up liquid: alkali. Pure water: neutral. [1 each]                                                                                                                                                                                                                                                                           | 3     |
|   | (c)  | Neutralisation                                                                                                                                                                                                                                                                                                                                        | 1     |
| 6 | (a)  | A composite material is made from two or more different materials combined to make a material with better properties than either alone [1]. Example: concrete (cement + gravel + sand + water) — strong under compression [1]. <i>Accept: fibreglass, reinforced concrete, wood (cellulose + lignin).</i>                                             | 3     |
|   | (b)  | Electrical wiring: copper; good electrical conductor. Disposable hot cup: polystyrene; good thermal insulator. Kitchen knife: stainless steel; hard, does not rust, keeps a sharp edge. Magnifying lens: glass; transparent, can be shaped to refract light. Chair seat: wood; strong, rigid, easy to shape. [1 per correct material + property pair] | 5     |
| 7 | (a)  | A=Battery (cell), B=Switch, C=Bulb (lamp) 1, D=Bulb (lamp) 2, E=Ammeter. [1 each]                                                                                                                                                                                                                                                                     | 5     |
|   | (b)  | The bulbs will not light [1]; an open switch breaks the circuit so current cannot flow [1].                                                                                                                                                                                                                                                           | 2     |
|   | (c)  | Bulb D would also go out [1]; in a series circuit there is only one path for the current, so if one component breaks the circuit is broken and no current flows through any component [1].                                                                                                                                                            | 2     |
| 8 | (a)  | Any two pairs: Series — only one path for current / all components share the same current / if one fails all fail / voltage is shared. Parallel — multiple paths for current / components can                                                                                                                                                         | 4     |



| Q        | Part    | Acceptable Answer(s)                                                                                                                                                                                                                                                     | Marks |
|----------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|          |         | operate independently / same voltage across each branch. [2 per clear difference, max 4]                                                                                                                                                                                 |       |
|          | (b)     | In parallel each bulb has its own path, so if one bulb fails the others stay on [1]; also each bulb gets the full voltage so they shine brightly [1].                                                                                                                    | 2     |
|          | (c)     | Any three from: copper, silver, gold, aluminium, iron, steel, graphite/carbon. [1 each]                                                                                                                                                                                  | 3     |
| 9        | (a)     | Any three from: do not touch sockets with wet hands; do not overload sockets; do not poke objects into sockets; switch off appliances when not in use; do not use damaged cables. [1 each]                                                                               | 3     |
|          | (b)     | The fuse contains a thin wire that melts if the current rises above a safe level [1], breaking the circuit and preventing damage to the appliance or a fire [1].                                                                                                         | 2     |
|          | (c)     | Wind wire around an iron nail many times; connect wire ends to battery terminals; the nail becomes a magnet when current flows [1 per step, max 1 for method]. Use: electric motor, door bell, crane, relay switch [1 for any valid use].                                | 2     |
| 10       | (a)     | Weather: the day-to-day conditions of the atmosphere (temperature, rainfall, wind). Climate: the average weather pattern of a place over many years. [1 each]                                                                                                            | 2     |
|          | (b)(i)  | Curepipe [1]; it is at a higher altitude (it is in the central plateau/uplands), so temperatures are cooler and clouds drop more rainfall on the windward side of the hills [1].                                                                                         | 2     |
|          | (b)(ii) | $28 - 22 = 6^{\circ}\text{C}$                                                                                                                                                                                                                                            | 1     |
|          | (c)     | Cyclone = a powerful tropical storm with very strong winds rotating around a low-pressure centre [1]. Effects: flooding; destruction of buildings and crops; strong winds uproot trees; power cuts; disruption to transport. [1 per effect, max 2]                       | 3     |
| 10 cont. | (d)     | Any two: burning fossil fuels (releases $\text{CO}_2$ ); deforestation (removes trees that absorb $\text{CO}_2$ ); farming (methane from livestock); industrial processes; transport. [1 each]                                                                           | 2     |
| 11       | (a)     | Tropical: hot and wet all year. Temperate: warm summers and cool winters, moderate rainfall. Polar: very cold, ice and snow, little rainfall. [1 per correct zone + characteristic pair, max 3]                                                                          | 3     |
|          | (b)     | The Sun's energy passes through the atmosphere and warms the Earth's surface [1]; the Earth releases infrared radiation (heat) [1]; greenhouse gases in the atmosphere trap some of this heat and prevent it escaping into space [1], keeping the Earth warm [1]. Excess | 4     |

| Q            | Part | Acceptable Answer(s)                                                                                                                         | Marks      |
|--------------|------|----------------------------------------------------------------------------------------------------------------------------------------------|------------|
|              |      | greenhouse gases trap too much heat, causing global warming. [max 4]                                                                         |            |
|              | (c)  | Any two from: solar energy, wind energy, hydroelectric power, wave/tidal energy, geothermal energy. [1 each]                                 | 2          |
|              | (d)  | Any one: flooding of low-lying coastal areas; loss of beaches; saltwater intrusion into freshwater; displacement of coastal communities. [1] | 1          |
| <b>TOTAL</b> |      |                                                                                                                                              | <b>100</b> |